

The following are a few of the prominent cloud based platforms that can be used for advanced implementations including data science, data exploration, machine learning, deep learning, artificial intelligence, etc.

Neptune

URL: <https://neptune.ml/>

Neptune is a lightweight cloud based service for high performance applications including data science, machine learning, predictive knowledge discovery, deep learning, modelling training curves and many others. Neptune can be integrated with Jupyter notebooks so that Python programs can be easily executed for multiple applications.

The Neptune dashboard is available at <https://ui.neptune.ml/> on which multiple experiments can be performed. Neptune works as a machine learning lab on which assorted algorithms can be programmed and their outcomes can be visualised. The platform is available as Software as a Service (SaaS) so that the deployment can be done on the cloud. The deployments can be done on the users' own hardware and can be mapped with the Neptune cloud.

In addition to having a pre-built cloud based platform, Neptune can be integrated with Python and R programming so that high performance applications can be programmed. Python and R are prominent programming environments for data science, machine learning, deep learning, Big Data and many other applications.

For Python programming, Neptune provides neptune-client so that communication with the Neptune server can be achieved, and advanced data analytics can be implemented on its advanced cloud.

For integration of Neptune with R, there is an amazing and effective library 'reticulate' which integrates the use of neptune-client.

The detailed documentation for the integration of R and Python with Neptune is available at <https://docs.neptune.ml/python-api.html> and <https://docs.neptune.ml/r-support.html>.

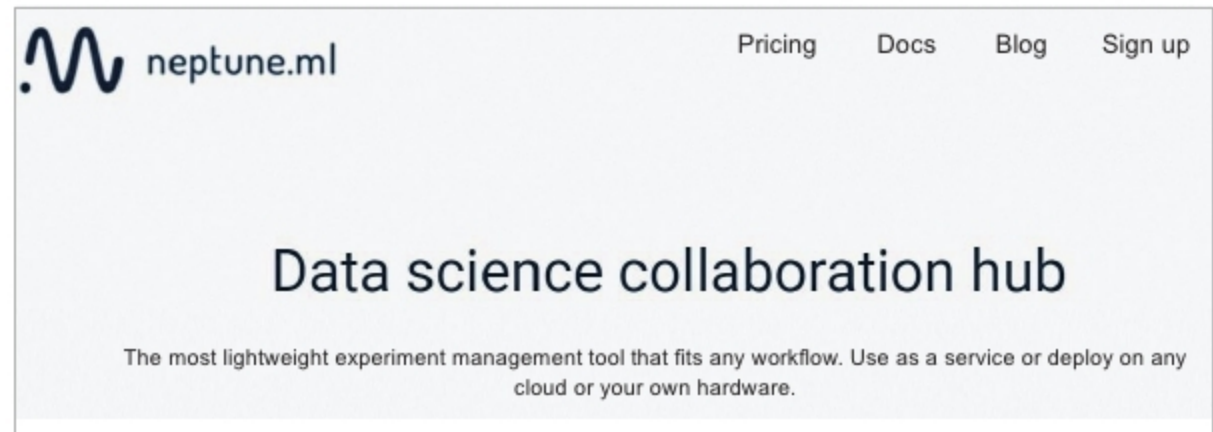


Figure 1: The Neptune portal

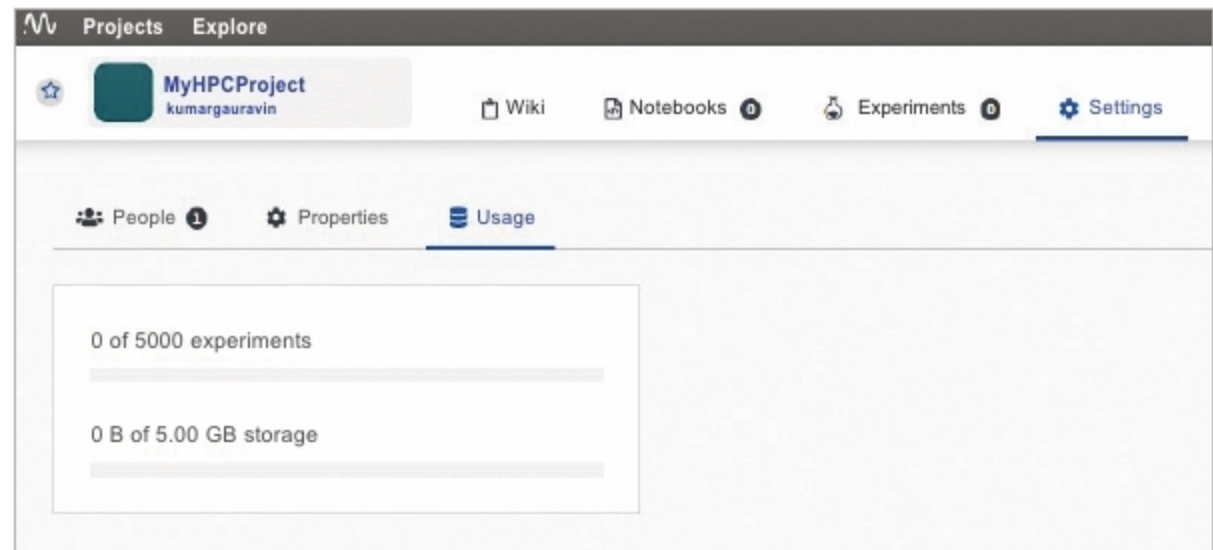


Figure 2: Creating a new project on the Neptune platform

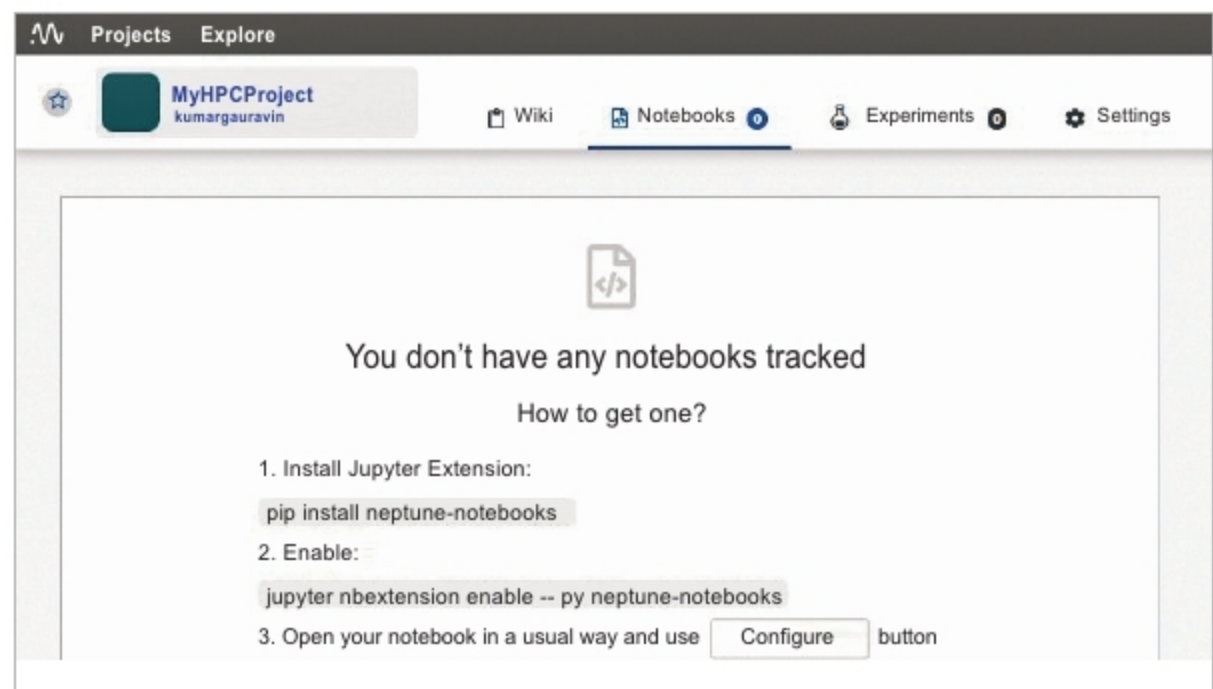


Figure 3: Integration of Neptune with Jupyter Notebook

[neptune.ml/python-api.html](https://docs.neptune.ml/python-api.html) and <https://docs.neptune.ml/r-support.html>.

In addition, integration with MLflow and TensorBoard is also available. MLflow is an open source platform for managing the machine learning life cycle with reproducibility, advanced experiments and deployments. It has three key components — tracking, projects and models. These can be programmed and controlled using the

Neptune - MLflow integration.

The association of TensorFlow with Neptune is possible using Neptune-TensorBoard. TensorFlow is one of the powerful frameworks for the deep learning and advanced knowledge discovery approaches.

With the use of assorted features and dimensions, the Neptune cloud can be used for high performance research based implementations.